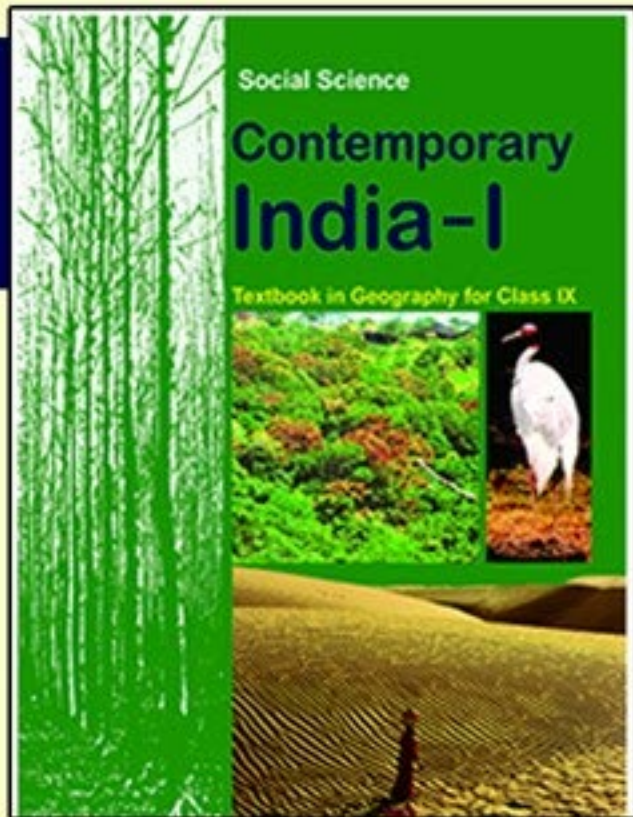




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NCERT BASICS



GEOGRAPHY

- ✓ MINERALS AND ENERGY RESOURCES
- ✓ INDUSTRIES

MODULE - 11





MINERALS



WHAT IS A MINERAL?

- A naturally occurring substance that has a definite chemical composition and physical properties is a **Mineral**.
- Minerals are formed in different types of geological environments, under varying conditions.
- They are created by natural processes without any human interference.
- They can be identified on the basis of their physical properties such as colour, density, hardness and chemical property such as solubility.



CLASSIFICATION OF**MINERALS**

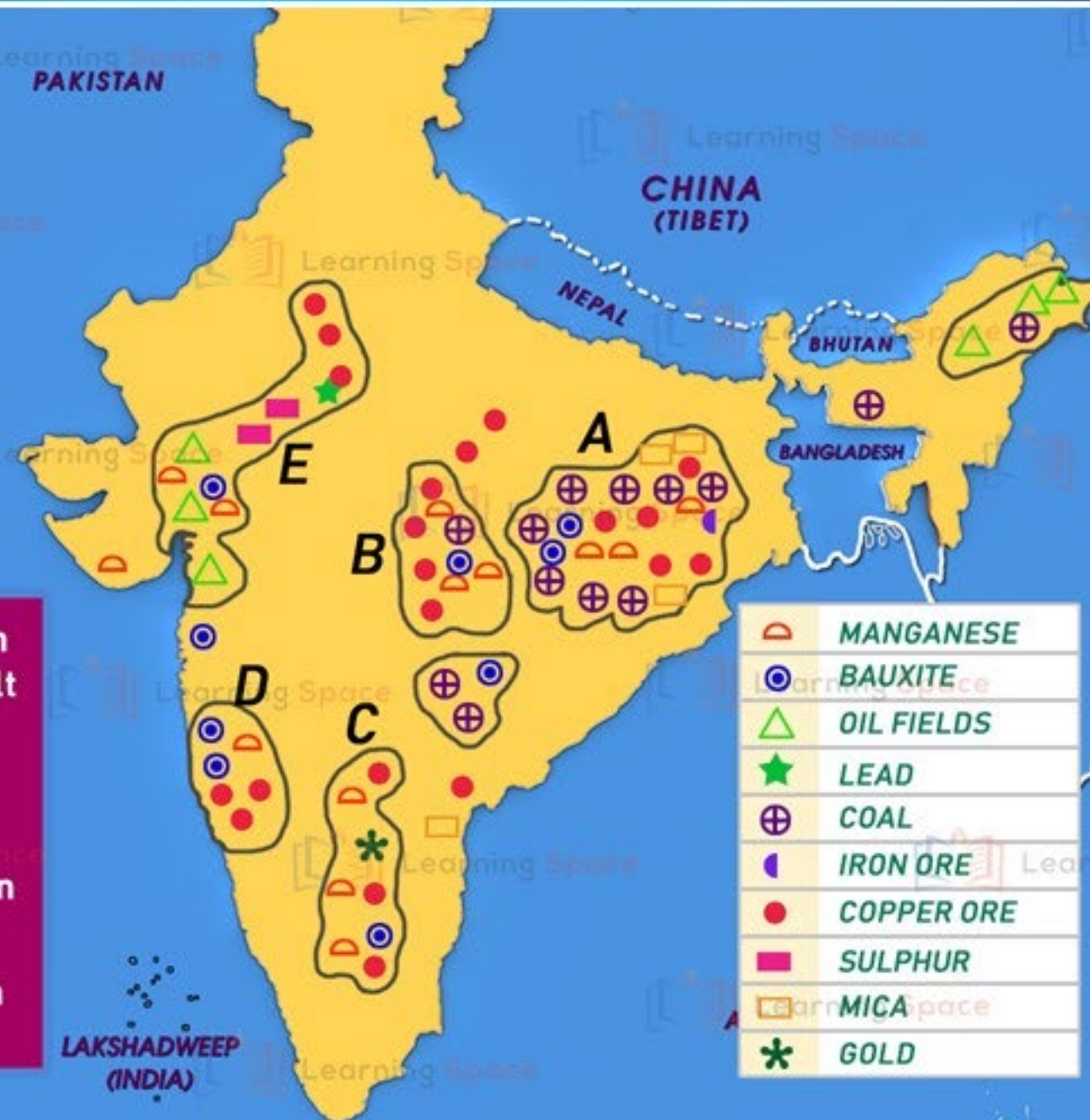
- Metallic minerals are the sources of metals.
- Non-metallic minerals are either organic in origin such as fossil fuels also known as mineral fuels which are derived from the buried animal and plant life such as coal and petroleum.
- Other type of non-metallic minerals are inorganic in origin.
- There is inverse relationship in quality and quantity of minerals.
- All minerals are exhaustible over time.

DISTRIBUTION OF MINERAL RESOURCES IN WORLD

MINERALS	COUNTRIES IN DECREASING ORDER OF PRODUCTION
Iron ore	China, Australia, Brazil, India, Russia
Maganese	China, South Africa, Australia, Brazil, Gabon
Bauxite	Australia, China, Brazil
Aluminium	China, Russia, Canada, UAE, India
Copper	Chile, China, Peru, USA, Congo
Silver	Mexico, China, Peru, Russia, Australia
Gold	China, Australia, Russia, USA, Canada
Tin	China, Indonesia, Peru, Bolivia, Brazil
Zinc	China, Peru, Australia, USA, Canada
Uranium	Kazakhstan, Canada, Australia, Niger

DISTRIBUTION OF IMPORTANT MINERAL AND ENERGY RESOURCES IN INDIA

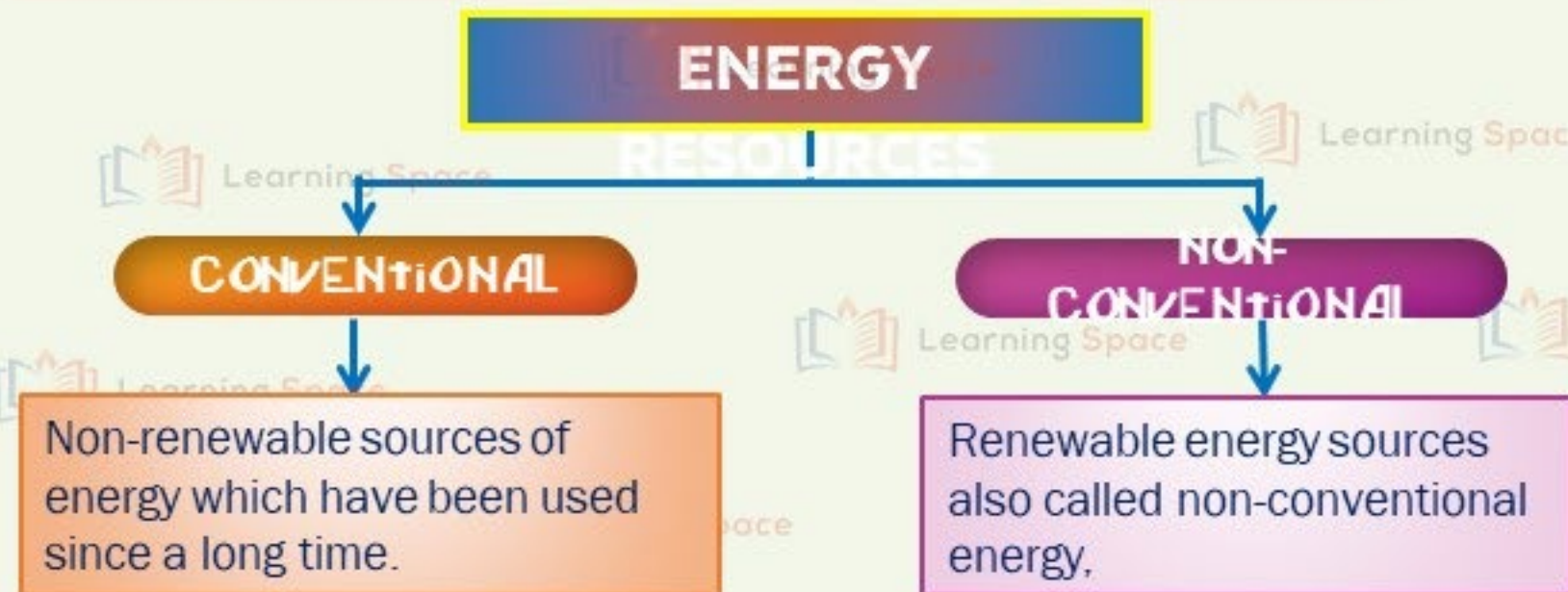
- A. North-Eastern Peninsular Belt
- B. Central Belt
- C. Southern Belt
- D. South-western Belt
- E. North-western Belt



	MANGANESE
	BAUXITE
	OIL FIELDS
	LEAD
	COAL
	IRON ORE
	COPPER ORE
	SULPHUR
	MICA
	GOLD

LET'S UNDERSTAND ABOUT ABOUT ENERGY RESOURCES!

- ✓ An energy resource is something that can produce heat, power life, move objects, or produce electricity.
- ✓ Energy can be generated from fuel minerals like coal, petroleum, natural gas, uranium and from electricity.
- ✓ Energy resources can be classified into 2 sources.



MINERALS AND ENERGY RESOURCES



Oil

Natural Gas

Advantages

- Easier to transport (tankers)
- Basis of Petro-chemical industry

Disadvantages

- Depletion of oxygen due to oil spilling and gas leakage
- Pollutants released caused acid rain
- Exploration of new fuel is not easy

Advantages

- Easier to transport (Pipelines)
- Cleaner than oil and coal
- Cheaper than oil



CONVENTIONAL SOURCES OF ENERGY

Firewood

Advantages

- Easy access
- Provides energy to a large number of people

Disadvantages

- Collection is time consuming
- Polluting
- Promoting green house effect
- Deforestation



Coal

Advantages

- Extensively available
- Efficient conversion to electricity

Disadvantages

- Polluting source
- Bulky to transport



Hydel Power

Advantages

- Non-polluting
- Promotes irrigation and fishing
- Cheap

Disadvantages

- Displacement of local community
- Inundates low lying regions
- Expensive to setup





Wind Energy

Advantages

- Non-polluting
- Low-cost production of electricity once setup
- Safe and clean

Disadvantages

- Noise pollution
- Windmills costly to setup
- Disturbs radio and T.V. reception
- Harmful to birds



Solar Energy

Advantages

- Inexhaustible
- Nonpolluting

Disadvantages

- Expensive
- Diffuse source, so gets wasted

Tidal Energy

Advantages

- Non-polluting
- Inexhaustible

Disadvantages

- Destroys wildlife habitat
- Difficult to harness

NON-CONVENTIONAL SOURCES OF ENERGY

Nuclear Energy

Advantages

- Emits large amount of energy

Disadvantages

- Generates radioactive wastes
- Expensive



Bio Gas

Advantages

- Low Cost
- Easy to operate
- Makes use of bio waste

Disadvantages

Causes green house effect

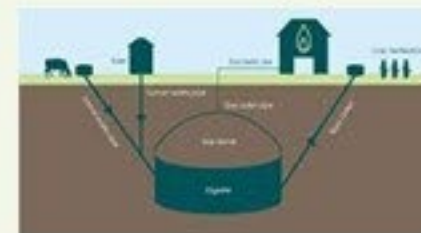
Geothermal Energy

Advantages

Clean ecofriendly and always available

Disadvantages

Located far away from cities and so costly to transport the electricity



DISTRIBUTION OF ENERGY RESOURCES IN WORLD

Energy resources are generally defined as anything that can be used as a source of energy.

RESOURCES	MAJOR PRODUCER COUNTRIES
Coal	China, USA, Australia, India
Petroleum	USA, Saudi Arabia, Russia, China
Natural Gas	USA, Russia, Qatar, Iran
Uranium	Kazakhstan, Canada, Australia, Niger
Wind Energy	China, USA, Germany, Spain, India
Solar Energy	Germany, China, Italy, Japan, USA, Spain
Hydro-electricity	China, Canada, Brazil, USA, Norway, New Zealand, India

DISTRIBUTION OF IMPORTANT MINERAL AND ENERGY RESOURCES IN INDIA

- A. North-Eastern Peninsular Belt
- B. Central Belt
- C. Southern Belt
- D. South-western Belt
- E. North-western Belt





- Secondary activities add value to natural resources by transforming raw materials into valuable products.
- Secondary activities, are concerned with manufacturing, processing and construction (infrastructure) industries.



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CLASSIFICATION OF

INDUSTRIES

Size**Inputs/Raw Materials****Output/Product****Ownership**

- ✓ Cottage or Household
- ✓ Small Scale
- ✓ Large Scale

- Agro based
- Mineral based
- Chemical based
- Forest based
- Animal based
- Knowledge based

- ✓ Basic
- ✓ Consumer Goods

- Public Sector
- Private Sector
- Joint Sector





CLASSIFICATION OF INDUSTRIES

- Another common classification of industries is based on the nature of the manufactured products.
- Eight classes of industries, thus identified are:
 - (1) Metallurgical Industries
 - (2) Mechanical Engineering Industries
 - (3) Chemical and Allied Industries
 - (4) Textile Industries
 - (5) Food Processing Industries
 - (6) Electricity Generation
 - (7) Electronics
 - (8) Communication Industries.



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MAJOR INDUSTRIES IN INDIA

IMPORTANT INDUSTRIES OF INDIA

- The Indian Government has been trying to promote rapid industrial growth since Independence.
- Some of the important industries in India are:
 1. Iron and Steel Industry
 2. Cotton and Textile Industry
 3. Cement Industry
 4. Petroleum and Natural Gas



MAJOR INDUSTRIES IN INDIA

1. IRON AND STEEL INDUSTRY

- First Steel industry was set-up at Kulti (West Bengal) under Bengal Iron Works Company in 1870.
- First large scale steel plant TISCO was set-up at Jamshedpur in 1907 followed by IISCO at Burnpur in 1919.
- The first public sector unit was 'Visveshvaraya Iron and Steel Works' at Bhadravati.
- The Ministry of Steel in India helps in planning and making sustainable growth and development in this industry.

INDUSTRIES

PUBLIC SECTOR STEEL PLANTS LOCATION	ASSISTANCE
Rourkela (Odisha)	Germany
Bhilai (Chhattisgarh)	Russian Government
Durgapur (West Bengal)	Britain Government
Bokaro (Jharkhand)	Russian Government
Burnpur (West Bengal)	Acquired by Private Sector in 1976
Visakhapatnam (Andhra Pradesh)	Russian Government
Bhadravati (Karnataka)	Nationalisation of Vishveshvarayya Iron and Steel Limited (owned by Central and State Government)

DISTRIBUTION OF IRON AND STEEL PLANTS IN INDIA

IRON AND STEEL PLANTS





MAJOR INDUSTRIES IN INDIA

2. COTTON AND TEXTILE INDUSTRY

- ✓ Oldest industry of India and employs largest number of workers.
- ✓ It is the largest organised and broad-based industry.
- ✓ The first Indian modernised cotton cloth mill was established in 1818 at Fort Gloaster near Calcutta, but this mill was not successful.
- ✓ The second mill named Bombay Spinning and Weaving Company was established in 1854 at Bombay by KGN Daber.



DISTRIBUTION OF COTTON TEXTILE INDUSTRIES IN INDIA

COTTON TEXTILE INDUSTRIES



3. CEMENT INDUSTRY

- ✓ Production of cement was started in 1904 at Madras, but the foundation of stable Indian cement industry was laid in 1914 when the Indian Cement Company Limited started production at Porbander in Gujarat.
- ✓ During planning period, the cement industry has recorded continuous growth.
- ✓ By the end of Seventh Plan, India started the export of cement.



4. PETROLEUM AND NATURAL GAS

- First successful oil well was dug in India in 1889, at Digboi, Assam.
- At present a number of regions having oil reserves have been identified and oil is being extracted in these regions.
- For exploration purpose, Oil and Natural Gas Commission (ONGC) was established in 1956 at Dehradun and Uttarakhand.





LOCATION OF INDUSTRIES

- This is influenced by several factors like:
 - Access To Raw Materials
 - Power
 - Market
 - Capital
 - Transport
 - Labour, etc.
- It is economical to locate the manufacturing industries at a place where cost of production and delivery cost of manufactured goods to consumers are the least.
- Transport costs, to a great extent, depend on the nature of raw materials and manufactured products.



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INDUSTRIES OF WORLD AND THEIR DISTRIBUTION

The secondary activities add value to natural resources by transforming raw materials into valuable products

INDUSTRIES	MAJOR COUNTRIES
Iron and Steel	Russia, China, USA, Japan, India
Cotton	China, India, USA
Jute	India, Bangladesh, China
Woolen Textile	Australia, China, USA
Paper	China, USA, Germany, Japan
Ship Building	China, South Korea, Japan
Aircraft Building	USA, Britain, Russia, France
Automobile	China, USA, Japan, Germany, South Korea, India
Petro Chemical	USA, Belgium, Japan





INDUSTRIAL POLICIES

- Industrial Policy can be defined as a statement stating:
 - ✓ The role of government in industrial development.
 - ✓ The position of private and public sectors in the industrialization of the country.
 - ✓ The comparative role of large and small industries.
- It enlists rules and procedures that will monitor the growth and pattern of industrial activity.

- There are 3 important Industrial Policy Resolutions.
 - Industrial Policy Resolutions, 1948
 - Industrial Policy Resolutions, 1956
 - Industrial Policy Resolutions, 1991



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INDUSTRIAL POLICIES RESOLUTIONS 1991

MAJOR OBJECTIVES

- To build on the gains already made.
- Correct the distortions or weaknesses that have crept in.
- Maintain a sustained growth in productivity and gainful employment.
- Attain international competitiveness.

MEASURES INITIATED

- Abolition of industrial licensing
- Free entry to foreign technology
- Foreign investment policy
- Access to capital market
- Open trade
- Abolition of phased manufacturing programme.
- Liberalised industrial location programme.

Industrial Policy





INDUSTRIAL POLICIES RESOLUTIONS 1991

LIBERALISATION, PRIVATISATION, GLOBALISATION (LPG) AND INDUSTRIAL DEVELOPMENT IN INDIA

- ✓ The industrial policy has been liberalized to attract private investor both domestic and multi-nationals.
- ✓ New sectors like, mining, telecommunications, highway construction and management have been thrown open to private companies.
- ✓ Globalisation means integrating the economy of the country with the world economy.
- ✓ Under this process, goods and services along with capital, labour and resources can move freely from one nation to another.
- ✓ The thrust of globalisation has been to increase the domestic and external competition through extensive application of market mechanism and facilitating dynamic relationship with the foreign investors and suppliers of technology.



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